MILITARY SPECIFICATION

CHAIN ASSEMBLY, SINGLE LEG, 10,000 POUND CAPACITY, TYPE MA-1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a chain assembly to be used with the type MC-1 cargo tiedown.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	-	Box, Shipping, Fiberboard
PPP-B-676	-	Boxes, Setup

MILITARY

MIL-P-116	-	Preservation, Method of
DOD-D-1000	-	Drawing, Engineering and Associated List
MIL-H-6875	-	Heat Treatment of Steels (Aircraft Practice), Process for
MIL-T-8652	-	Tiedown, Cargo, Aircraft, 5,000 Lb Capacity, Type MC-1

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-143	-	Standards and Specifications Order of Precedence for the Selection of
MIL-STD-794	-	Part and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	-	Environmental Test Methods
MIL-STD-831	-	Test Reports, Preparation of
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment

* (Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the document which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

* ASTM A 304-78	-	Steel, Sheet, Cold Rolled, Long Terne Coated
* ANSI A 322-81	-	Steel Bars, Alloy, Standard Grades
* ASTM A 331-81	-	Steel Bars, Alloy, Cold Finished

* (Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* (Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The chain assembly furnished under this specification shall be a product which has been tested, and passed the qualification test specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Components. The chain assembly shall consist of a length of chain, one quick attach-detach (QAD) fitting, and one hook.

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3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3.1 AN or MIL standard parts. AN or MIL standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they possess suitable properties and are replaceable by the AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding AN or MIL standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.4 Materials.

3.4.1 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt, or spray, or atmospheric conditions likely to be met in storage or normal service. The chain shall be made of an alloy steel conforming to ASTM A 304, ASTM A 322, and ASTM A 331, or equivalent, heat treated to produce a chain of high tensile strength and resistance to impact loads.

3.4.1.1 Heat treatment. Heat treatment of steel parts shall be in accordance with MIL-H-6875.

3.4.1.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.5 Design and construction. The chain assembly shall be designed to have an ultimate strength of not less than 14,100 pounds (62716.8 n) and shall be so constructed that it forms a single unit without detachable parts. The chain assembly shall conform to the envelope shown on figure 1.

3.5.1 QAD fitting. The QAD fitting shall be designed for use with type MC-1 cargo tie downs conforming to MIL-T-8652. The QAD fitting shall include a feature, manually operable without the use of any tools, which will permit the easy attachment or detachment of the 1 3/4 inches (4.445 cm) wide webbing without removal of the webbing from the type MC-1 cargo tie down attachment fittings. The QAD fitting shall include a metal roller designed to reduce webbing friction when tightening the type MC-1 cargo tie down. No lubrication shall be required. The QAD fitting shall not disengage under load or no-load conditions unless manually actuated. The QAD fitting shall be attached to the coupling link of the chain in such a manner that the strength of neither the chain nor the QAD fitting shall be reduced.

3.5.2 Chain. The chain shall be constructed using the electric welding process. The dimensions of each link shall be as specified in the subparagraphs of 3.8.2, except the links at the ends of the chain may be of the next larger nominal trade size for the purpose of attaching to the QAD fitting and the hook.

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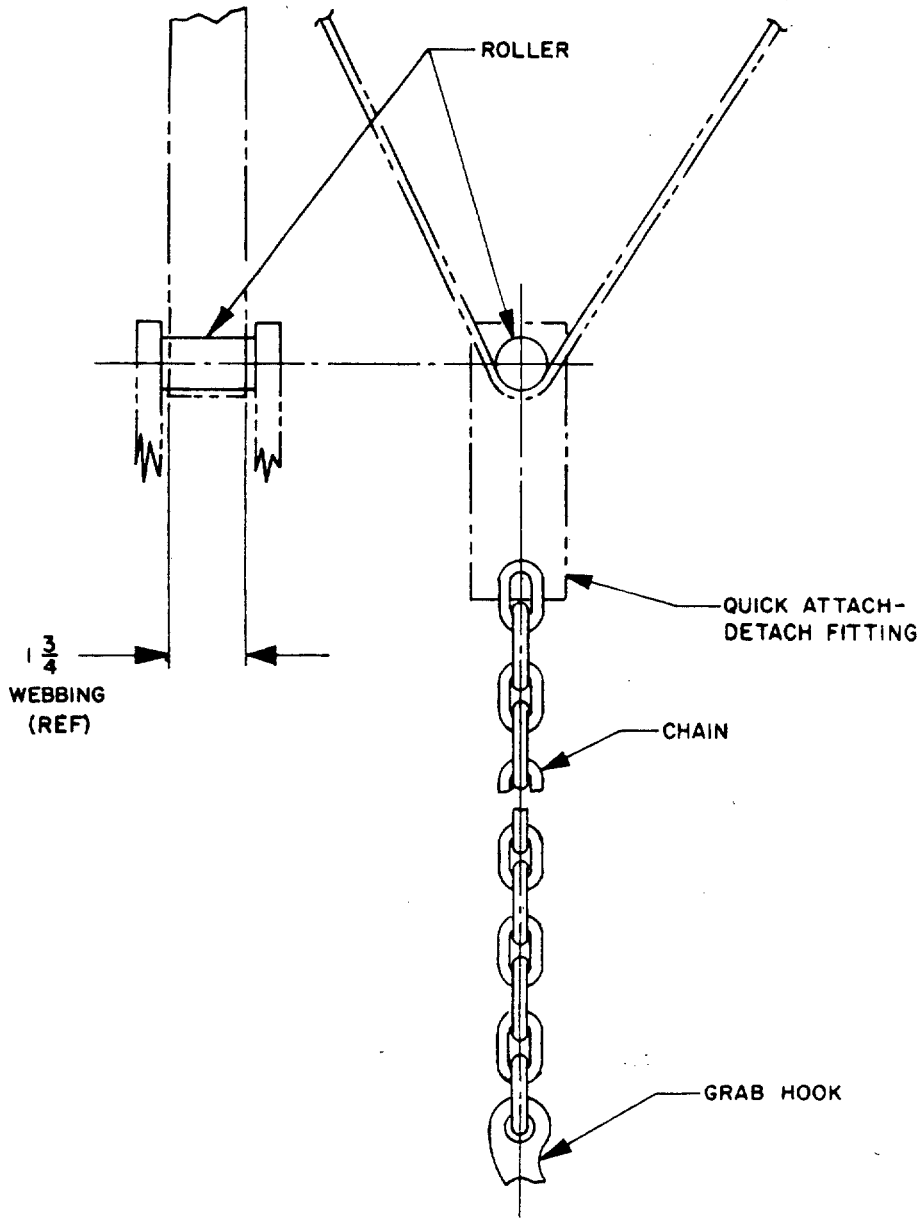


FIGURE 1. Chain assembly.

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3.5.3 Hook. The hook shall be a grab-type of suitable dimensions to engage chain links of the size specified in the subparagraphs of 3.8.2. When used in this manner, the hook shall meet the strength requirements as specified in 3.5, and the hook shall not reduce the strength of the chain assembly into which the hook has been attached. The hook shall be so designed that the chain assembly can be used in various combinations with chain assemblies having the same link dimensions. When used in this manner, the hook shall meet the strength as specified in 3.5, and the hook shall not reduce the strength of the chain assembly into which the hook has been attached.

3.6 Performance.

3.6.1 Proof load. When tested as specified in 4.6.2, the chain assembly shall withstand a proof load of 10,000 pounds (44.480 n) without any evidence of failure, permanent deformation, or subsequent malfunction of parts.

3.6.2 Ultimate load. When tested as specified in 4.6.3, the chain assembly shall sustain an ultimate load of 14,100 pounds (62716.8 n). Deformation, without rupture or visible cracks, will be permitted provided it does not prevent operation of the release mechanism of the QAD fitting.

3.6.3 Environmental characteristics. The chain assembly shall operate satisfactorily when subjected to the salt spray test in 4.6.4 and the low temperature test in 4.6.5.

3.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-D-1000 shall govern the manufacturer's part number and changes thereto.

3.8 Dimensions.

3.8.1 Length. The effective length of the chain assembly shall be 24 ±1 inch (60.96 ±2.54 cm).

3.8.2 Links.

3.8.2.1 The maximum outside length of the links shall be 1.521 inches (3.863 cm) after applying the ultimate load.

3.8.2.2 The maximum outside width of the links, including any weld buildup, shall be 0.975 inch (2.477 cm) after applying the ultimate load.

3.8.2.3 The maximum actual bar size of the links shall be 9/32 inch (.714 cm).

3.8.2.4 The minimum inside length of the links shall be 0.825 inch (2.096 cm) after applying the ultimate load.

3.9 Weight. The maximum weight of the chain assembly shall be 3 pounds (1.36 kg).

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3.10 Finish. All steel parts shall be treated with a metallic-type coating which will provide cathodic protection to the base metal. This protective coating shall not crack, peel, or flake during normal usage, and shall withstand any adverse conditions to which the chain assembly will be subjected in accordance with this specification. The protective coating shall in no way prevent compliance with the performance requirements of this specification.

3.11 Identification of product. Each chain assembly shall be marked for identification in accordance with MIL-STD-130. Markings shall be applied to the hook portion of the assembly or on a metal tag securely attached to the chain.

3.12 Workmanship. The workmanship shall be in accordance with high quality manufacturing practices common to the type of equipment. The chain assembly shall be free from injurious imperfections that may impair appearance or serviceability. The forged hook, welded chain and attachments shall be free from fins, cracks, flaws, seams, burrs, and sharp edges which may tend to cut or damage the webbing, or cause binding of the QAD fitting.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of chain assemblies shall be classified as follows:

a. Qualification tests.

b. Acceptance tests.

4.3 Test conditions. Unless otherwise specified, tests shall be conducted at normal atmospheric conditions.

4.4 Qualification testing.

4.4.1 Test sample. The test sample shall consist of two chain assemblies representative of the production equipment. The sample shall be identified by the manufacturer's part number and such other information as required by the procuring activity (see 6.3).

4.4.2 Test report. Upon completion of qualification testing, the contractor shall submit a test report prepared in accordance with MIL-STD-831.

4.4.3 Qualification tests. Qualification tests shall consist of all the tests specified in 4.6.

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4.5 Acceptance tests. Acceptance tests shall consist of examination of product and periodic sampling tests.

4.5.1 Periodic sampling tests. Sampling for proof load test (see 4.6.2) and ultimate load test (see 4.6.3) shall be in accordance with MIL-STD-105, unless otherwise specified herein. The sample shall be one chain assembly. The inspection level shall be S-2 with an acceptable quality level of 2.5, expressed in defects per hundred units.

4.6 Test methods.

4.6.1 Examination of product. Each chain assembly shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Proof test. Each sample chain assembly shall be placed in a testing machine in such a manner that the load is applied in tension to the grab hook, chain, and QAD fitting. After removal of the load, the link dimensions shall be as specified in the subparagraphs of 3.8.2 and the chain shall remain flexible. The chain shall then be passed around a 3 inch (7.62 cm) diameter bar and the hook fastened into a link, maintaining an included angle of 30 $\pm$5 degrees. The specified load shall again be applied. Upon release of the load, the chain assembly shall be examined for conformance to the requirement specified in 3.6.1.

4.6.3 Ultimate load test. Each sample chain assembly shall be subjected to the ultimate load test in accordance with the procedure in 4.6.2 except that the requirements shall be as specified in 3.6.2.

4.6.4 Salt spray test. One qualification sample chain assembly shall be subjected to the salt spray test in accordance with MIL-STD-810. The chain shall meet the operation requirements specified in Section 3.

4.6.5 Low temperature test. One chain assembly shall be subjected to the low temperature test in accordance with MIL-STD-810. While at the low temperature, the chain assembly shall be tested for compliance with the operational requirements specified in Section 3.

4.7 Inspection of the preservation, packaging and marking for shipment and storage Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

* 5. PACKAGING

* 5.1 Preservation. Preservation shall be level A, C or Industrial as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified in the contract or order each chain assembly shall be preserved and packaged in accordance with MIL-P-116, method III, one each in a unit container conforming to PPP-B-676.

* 5.1.2 Level C. Preservation shall conform to the requirements of MIL-STD-794 for this level.

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* 5.1.3 Industrial. Preservation shall be in accordance with the requirements of MIL-STD-1188.

* 5.2 Packing shall be level A, B, C or Industrial as specified (see 6.2).

5.2.1 Level A. Chain assemblies preserved and packaged as specified in 5.1.1 shall be packed in overseas weather resistant type shipping containers conforming to PPP-B-636. As far as practicable, shipping containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each shipping container shall not exceed the weight limitation of the specification. Containers shall be closed and strapped in accordance with the specifidation and appendix thereto.

5.2.2 Level B. Chain assemblies preserved and packaged as specified in 5.1.1 shall be packed in domestic type shipping containers conforming to PPP-B-636. Shipping containers shall be of minimum cube and tare consistent with the protection required. As far as practicable, shipping containers shall be of uniform shape and size and contain identical quantities. The gross weight of each shipping container shall not exceed the weight limitation of the specification. Containers shall be closed and strapped in accordance with the specification and appendix thereto. Shipping containers shall meet the special requirements table of PPP-B-636, as applicable.

* 5.2.3 Level C. The packing shall conform to the requirements of MIL-STD-794 for this level.

* 5.2.4 Industrial. The packing shall be in accordance with the requirements of MIL-STD-1188.

5.3 Marking.

* 5.3.1 Levels A, B, and C. In addition to any special or other identification marking required by the contract (see 6.2), each unit pack intermediate and exterior container shall be marked in accordance with MIL-STD-129.

* 5.3.2 Industrial. Industrial marking shall be in accordance with the requirements of MIL-STD-1188.

6. NOTES

6.1 Intended use. Chain assemblies covered by this specification are intended for use with type MC-1 tie down assemblies in accordance with MIL-T-8652. The chain assembly can be used with either one or two type MC-1 web straps in a two-branch configuration or with one web strap in a single-branch configuration.

* 6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable levels of preservation, packaging and packing required (see 5.1, 5.2 and 5.3).

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* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is (San Antonio ALC/MMIRCC, Kelly Air Force Base TX 78241) and information pertaining to qualification of products may be obtained from that activity.

6.3.1 Drawings. When requested, the manufacturers shall submit engineering drawings and inspections in accordance with SD-6 (Provisions Governing Qualifications).

* 6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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